10 inch FPV drone build handover

Photographed components, references, compatibility findings and continuation brief

Prepared 4 September 2026
Build basis 10 inch quadcopter using the on hand 4S battery and photographed components
Purpose A self contained handover for continuing design, wiring, firmware setup and build validation in a separate chat

Handover summary

The core airframe and electronics are present, but the build is not ready for a direct plug together assembly. The main blockers are the undocumented LUX to Orqa harness pinout, exact ArduPilot firmware handling, the unvalidated 3110 motor and 10 by 5 tri blade propeller pairing, separate power for the Raspberry Pi, and the need to keep the digital Pi camera distinct from the analog pilot video system.

Selected power system 4S only for this build phase. The FC, ESC and VTX voltage ranges accept 4S. No 6S replacement is proposed.

Package contents

- 10-inch-FPV-drone-handover.docx — this reference and continuation document
- 10-inch-FPV-drone-component-crosscheck.xlsx — annotated inventory crosscheck with Present, Missing, Partial, Verify and Conditional sheets
- 10-inch-FPV-drone-handover-package.zip — both files in one portable archive

Current component register

Subsystem	Photographed component	State	Immediate note
Flight controller	Lumenier LUX F765 NDAA	Present	Exact ArduPilot target and microSD requirement matter
ESC	Orqa 3030 70A 3S to 6S BLHeli 32 4 in 1 V2.2	Present	Legacy BLHeli 32 unit, not current AM32 version
Frame	Rekon FPV Rekon10 Pro V2	Present	10 inch, 474 mm wheelbase
Motors	iFlight XING2 3110 900KV	Present	19 by 19 mm M3 mount
Propellers	Gemfan 1050 Cinelifter 10 by 5 tri blade	Present	5 mm hub, motor recommendation mismatch
Battery	Tattu G Tech 5200 mAh 4S 35C XT60	Present	4S is the selected build voltage
Receiver	Matek ELRS R24 D 2.4 GHz diversity	Present	CRSF on full UART
GNSS and compass	Matek M10Q 5883	Present	UART plus I2C
Video transmitter	Rush FPV Tank Solo analog 5.8 GHz	Present	Antenna required before power
Companion computer	Raspberry Pi Zero 2 W with 64 GB microSD	Present	Dedicated airborne 5 V regulator required
Companion camera	Raspberry Pi Camera Module 2	Present	CSI digital camera, not an analog FPV camera
Bench power	CanaKit DCAR 052A5 5 V 2.5 A AC supply	Present	Bench only, not an airborne LiPo regulator

Compatibility assessment

Priority	Topic	Finding	Required action
Critical	FC to ESC wiring	The matching 30.5 mm mounting pattern does not prove connector pin compatibility. The LUX kit includes 8 pin JST SH leads, while the Orqa ESC listing does not document a plug compatible harness.	Map motor outputs, battery voltage, current sense, ESC telemetry and grounds from both boards. Repin or solder to verified pads. Continuity test before connecting the LiPo.
Critical	ArduPilot target and boot media	Lumenier states that the LUX F765 requires a microSD card to boot after ArduPilot or PX4 is installed.	Use the exact LUXNDAAF7 firmware target and a known good FAT32 microSD. Do not substitute a generic F765 target.
High	ESC firmware identity	The photographed ESC is the older BLHeli 32 V2.2 product. Current Orqa ESC documentation describes an AM32 product and is not interchangeable firmware guidance.	Read the installed ESC identity through ArduPilot passthrough after wiring is verified. Do not cross flash AM32 firmware without an exact, official hardware target.
High	Motor and propeller pairing	Gemfan recommends 3212 to 3214 motors for the 1050 tri blade. The on hand motors are smaller 3110 units. Mechanical fit alone does not validate current, heat or thrust.	Perform a restrained one motor thrust and current test on 4S, then inspect temperatures and logs before a four motor test or flight.
High	Analog and digital video	The Raspberry Pi Camera Module 2 outputs CSI digital video and cannot feed the Rush analog VTX directly.	Use the separate analog FPV camera identified in the inventory for piloting. Treat the Pi camera as a recording or computer vision payload.
High	Pi power	The CanaKit supply accepts AC mains, not 4S LiPo input. The Rush 5 V camera output is far below a sensible Pi power budget.	Use a dedicated 4S input regulated 5 V BEC for the Pi. Budget at least 3 A and verify ripple, connector polarity and common ground.
High	Power system current	The 4S 5200 mAh 35C pack is nominally rated around 182 A continuous, while published motor peak figures cannot be transferred directly to this prop and voltage combination.	Treat the XT60, battery leads, ESC, solder joints and current sensor as one system. Use a smoke stopper only for electronics checks, not motor load tests. Log current during controlled testing.
Medium	VTX supply rail	The Tank Solo accepts 7 to 36 V and the LUX offers a switched 10 V VTX rail, but the available page does not establish the rail current margin for maximum VTX output.	Confirm the LUX manual current rating. Use the 10 V rail only with adequate margin; otherwise use filtered battery voltage. Never power the VTX without an antenna.
Medium	2.4 GHz coexistence	The Pi Zero 2 W Wi Fi and the ELRS receiver share the 2.4 GHz band.	Separate antennas and wiring. Consider disabling Pi Wi Fi in flight or testing coexistence at intended ELRS power and range.
Medium	GNSS compass interference	The external compass can be disturbed by motors, ESC current, battery leads, VTX and ferrous hardware.	Mount the module at least 10 cm from high current wiring and propulsion hardware when practical, then calibrate and verify compass consistency.
Medium	Propeller hardware	The official iFlight page describes a 5 mm shaft and M5 nut, but verify the actual motors because distributor descriptions can vary.	Dry fit a prop and verify shaft diameter, thread direction and nut engagement before ordering or installing locknuts.
Conditional	Remote ID	No Remote ID module was identified in the inventory.	Add an FAA compliant module only if the intended operation requires it.

Component references

These links are the primary operating references located for the exact products or the closest authoritative product documentation. Packaging text is treated as identification, not as an instruction to alter firmware or wiring.

Lumenier LUX F765 flight controller

Photographed identification Lumenier LUX F765 Flight Controller NDAA, retailer SKU 22726

- STM32F765 flight controller with 30.5 mm mounting, eight UARTs, twelve outputs, microSD slot and regulated accessory rails.
- Official product information lists Betaflight, ArduPilot and PX4 support.
- Lumenier explicitly states that a microSD card is required to boot when using ArduPilot or PX4.
- The package normally includes two 8 pin JST SH cables, but cable presence does not establish Orqa ESC pin compatibility.

Integration notes Use only the exact LUXNDAAF7 ArduPilot target. The Lumenier download marked with bootloader is a DFU first flash image; later Mission Planner custom updates normally use the target specific ArduPilot package. Preserve a known working firmware copy before changing versions.

Official product page and downloads https://www.lumenier.com/products/lumenier-lux-f765-flight-controller-ndaa

Lumenier NDAA declarations https://www.lumenier.com/military-and-defense-drones/ndaa-declarations

Orqa 3030 70A BLHeli 32 ESC

Photographed identification Orqa 3030 70A 3S to 6S BLHeli 32 4 in 1 ESC NDAA, model G24083HR1Z, package revision V2.2

- Four channel 70 A continuous ESC, 80 A for 10 seconds, 30 by 30 mm mounting and DShot support.
- This is the legacy BLHeli 32 version shown in the photo, not the current Orqa AM32 product.

Integration notes Do not apply AM32 firmware or assumptions from the current Orqa product page. First confirm board pinout and installed BLHeli 32 identity through a verified connection. Map every FC to ESC conductor before power.

Exact legacy product page https://www.lumenier.com/products/orqa-3030-70a-3-6s-blheli_32-4-in-1-esc-ndaa

Orqa manual portal https://orqafpv.com/manual

Current AM32 product for distinction only https://shop.orqafpv.com/products/orqa-orqa-electronic-speed-controller

Rush FPV Tank Solo video transmitter

Photographed identification Rush FPV Tank Solo analog 5.8 GHz VTX with MMCX to SMA cable

- Input range 7 to 36 V, 48 channels and selectable PIT, 25, 400, 800 and maximum output modes.
- Provides a 5 V camera output rated far below a Raspberry Pi power requirement.
- Uses analog video and SmartAudio control.

Integration notes Power from the LUX switched 10 V rail only after confirming regulator current margin; otherwise use filtered battery voltage. Install the antenna before applying power. The Pi Camera Module 2 cannot connect to this VTX.

Official product reference https://rushfpv.net/products/tank-solo-vtx

Raspberry Pi Zero 2 W

Photographed identification Raspberry Pi Zero 2 W with a photographed 64 GB microSD card

- Uses a 22 pin, 0.5 mm pitch mini CSI camera connector, microSD storage, mini HDMI and micro USB power.
- Official guidance recommends a 2 A supply for the board; attached peripherals add to the power budget.
- Onboard Wi Fi operates in the same 2.4 GHz band as the ELRS receiver.

Integration notes Use a dedicated 4S input 5 V regulator sized with margin, ideally at least 3 A for the Pi and camera. Do not power the Pi from the Rush camera output. Test ELRS range with Pi Wi Fi active and disabled.

Official product page https://www.raspberrypi.com/products/raspberry-pi-zero-2-w/?variant=raspberry-pi-zero-2-w

Official Raspberry Pi hardware documentation https://www.raspberrypi.com/documentation/computers/raspberry-pi.html

Official connector and radio hardware documentation https://www.raspberrypi.com/documentation/hardware/rf/

Raspberry Pi Camera Module 2

Photographed identification Raspberry Pi Camera Module 2, 8 megapixel Sony IMX219

- Fixed focus CSI digital camera supporting common still and video modes.
- A Pi Zero camera cable is required because the Pi Zero uses the narrow 22 pin connector while the camera board uses the standard camera connector.
- Current Raspberry Pi OS uses rpicam applications and Picamera2; the legacy raspivid and raspistill stack is unsupported.

Integration notes This camera is for recording, computer vision or a companion link. It is not a substitute for the analog FPV camera feeding the Rush VTX.

Camera hardware documentation https://www.raspberrypi.com/documentation/accessories/camera.html

Camera software documentation https://www.raspberrypi.com/documentation/computers/camera_software.html

Camera Module 2 documentation portal https://pip.raspberrypi.com/categories/652-raspberry-pi-camera-module-2

Matek ELRS R24 D receiver

Photographed identification Matek ELRS-R24-D 2.4 GHz diversity receiver, CRSF, 22.5 dBm telemetry

- Accepts 4 to 9 V and connects by CRSF using a full bidirectional UART.
- Receiver RX connects to FC TX and receiver TX connects to FC RX, with 5 V and ground.
- ExpressLRS 3 series firmware target is MATEK R24-D 2.4GHz RX.

Integration notes Use a compatible 2.4 GHz ELRS transmitter with matching major firmware, regulatory domain and binding phrase. Keep both antennas clear of carbon, high current wiring, VTX hardware and the Pi radio.

Official Matek product manual https://www.mateksys.com/?portfolio=elrs-r24

ExpressLRS receiver wiring https://www.expresslrs.org/quick-start/receivers/wiring-up/

ExpressLRS receiver binding https://www.expresslrs.org/quick-start/receivers/binding/

Matek M10Q 5883 GNSS and compass

Photographed identification Matek M10Q-5883 with u-blox M10 GNSS and QMC5883L compass

- Accepts 4 to 9 V and uses a six pin JST GH connection.
- GNSS uses a UART while the compass uses I2C clock and data lines.
- Matek specifies ArduPilot 4.3 or later and Rotation None for a flat ArduPilot installation.

Integration notes Cross TX and RX, connect SCL and SDA to the FC I2C bus, and keep the module at least 10 cm from high current conductors, ESC, motors and ferrous hardware where practical. Calibrate outdoors after the final installation.

Official Matek product manual https://www.mateksys.com/?portfolio=m10q-5883

ArduPilot compass calibration https://www.ardupilot.ardupilot.org/copter/docs/common-compass-calibration-in-mission-planner.html

iFlight XING2 3110 900KV motors

Photographed identification iFlight XING2 3110 900KV FPV cinelifter motors

- Official listings describe a 19 by 19 mm M3 mount, approximately 77 g mass, 18 AWG leads and a 5 mm prop shaft for the 900KV version.
- The official peak current and power data are test point figures and are not a safe prediction for this 4S and Gemfan propeller combination.

Integration notes The motor mount matches the Rekon10 frame. Verify the actual shaft diameter and included M5 hardware. The 3110 motor is smaller than Gemfan's 3212 to 3214 recommendation for the 1050 tri blade, so validation testing is mandatory.

Official iFlight EU product page https://iflight-rc.eu/en-gb/products/xing2-3110-fpv-cinelifter-motor?variant=45302443573513

Official iFlight thrust data search https://www.iflight.com/index.php?keyword=33&route=resources%2Ftechnical_support_search&section=support-2-19

Rekon10 Pro V2 frame

Photographed identification Rekon FPV Rekon10 Pro V2 10 inch frame

- Published specifications list a 474 mm wheelbase, 30.5 by 30.5 and 20 by 20 mm FC mounts, and 19 by 19 mm M3 motor mounts.
- VTX mounting patterns include 30.5, 20 and 25.5 mm; camera width is 19 to 20 mm.
- Published frame mass is approximately 567 g including hardware.

Integration notes No standalone official assembly manual was located. Dry fit the carbon stack, confirm arm and plate orientation from product images, isolate electronics from carbon edges, and use threadlocker only on metal to metal fasteners.

Detailed product dimensions and mounting reference https://www.fpv24.com/da/hglrc/hglrc-rekon10-pro-v2-long-range-ramme-10-tommer-o4-kompatibel

Tattu G Tech 5200 mAh 4S battery

Photographed identification Tattu G-Tech 5200 mAh 14.8 V 35C 4S1P with XT60, SKU TAA52004S35X6

- Published mass is approximately 436.5 g with dimensions about 133 by 45 by 33.5 mm.
- Uses an XT60 discharge connector, JST XHR balance connector and 10 AWG leads.
- Nominal energy is 76.96 Wh; the label discharge rating corresponds to about 182 A, but system limits may be lower.

Integration notes Confirm secure two strap retention, pad the pack from carbon, and keep balance leads away from propellers. Begin propulsion tests at low duty cycle and use current logging rather than treating the C rating as a guaranteed aircraft current limit.

Official Tattu product page https://genstattu.com/tattu-5200mah-14-8v-35c-4s1p-lipo-battery-pack-with-xt60-plug.html

Gemfan 1050 Cinelifter propellers

Photographed identification Gemfan 1050 10 by 5 glass fiber nylon tri blade set, one clockwise and one counterclockwise per pack

- Published mass is 16.8 g per propeller with a 5 mm shaft hole.
- Gemfan recommends 3212 to 3214 motors for this propeller.
- A quad requires two clockwise and two counterclockwise propellers plus practical spares.

Integration notes Do not install props during firmware, receiver, compass or motor output configuration. Confirm blade orientation and motor direction only after the ArduPilot motor order test. Balance or reject visibly damaged props.

Official Gemfan product page https://www.gemfanhobby.com/1050-cinelifter-glass-fiber-nylon-3-blade.html

CanaKit 5 V 2.5 A power supply

Photographed identification CanaKit model DCAR-052A5, 100 to 240 V AC input, 5 V 2.5 A output

- This is an indoor AC mains power adapter suitable for bench power when its connector and polarity match the Pi.
- It is not a 4S input BEC and must not be connected to the flight battery.

Integration notes Use for stationary Raspberry Pi setup only. The airborne Pi installation needs a separate regulated DC supply from the 4S power system.

UL Product iQ listing for model family https://productiq.ulprospector.com/en/profile/2664639/qqjq7.e471380?page=25&term=QQJQ7

ArduPilot and Mission Planner

Recommended firmware path Start with the Lumenier supplied LUXNDAAF7 ArduCopter image to establish a known board target and bootloader state. If the controller has no compatible bootloader, the Lumenier file named with_bl.hex is a DFU image. Once the correct bootloader is installed and Mission Planner recognizes the board, use only a confirmed LUXNDAAF7 ArduPilot package for later updates. Never select a generic F765 target merely because the processor matches.

Firmware sequence

1. Remove all propellers and disconnect the main battery. Photograph both boards and connector labels before assembly.
2. Install a known good FAT32 microSD in the LUX F765. Lumenier states that ArduPilot boot requires it.
3. Connect the FC by USB only. Record the current firmware and board identity before flashing.
4. For an initial DFU flash, use the exact Lumenier LUXNDAAF7 image that includes the bootloader. For a normal Mission Planner custom update, use the exact target specific ArduPilot firmware package.
5. After reboot, confirm Mission Planner connects, the board orientation is correct and the SD card is logging.
6. Wire only the receiver and GNSS next. Configure and verify each UART before adding ESC and VTX power wiring.
7. Wire the ESC only after a pin by pin map has been documented and checked by continuity meter.
8. Use DShot so throttle calibration is not required. Start with a conservative DShot rate if signal leads are long; confirm motor order and direction with props removed.
9. Back up parameters after each stable milestone and name files with the date and hardware state.

Mission Planner parameter starting points

Function	Starting point	Validation
Receiver UART	SERIALx_PROTOCOL = 23	CRSF and ExpressLRS require TX and RX on a full UART. Baud is normally detected automatically.
ESC protocol	MOT_PWM_TYPE = DShot option supported by the installed ArduPilot version	DShot requires no ESC throttle calibration. Choose a conservative rate first and follow the exact output group capability.
GNSS UART	GPS protocol auto detection initially	Verify satellites, fix type and update rate outdoors before arming.
Compass	External compass enabled, Rotation None for flat Matek mount	Confirm compass instance and orientation, then calibrate after final mounting.
Logging	SD logging enabled	Confirm log creation before motor load testing.
Frame	Quad X only after motor layout is verified	Use Mission Planner motor test at very low power with props removed.

Official ArduPilot references

Mission Planner firmware loading https://ardupilot.ardupilot.org/planner/docs/common-loading-firmware-onto-pixhawk.html

Mission Planner custom firmware https://ardupilot.org/planner/docs/common-custom-firmware.html

Mission Planner initial setup https://ardupilot.org/planner/docs/mission-planner-initial-setup.html?highlight=joystick

Mandatory hardware configuration https://ardupilot.org/copter/docs/configuring-hardware.html?highlight=mission

Motor order and motor test https://en.ardupilot.org/copter/docs/connect-escs-and-motors.html

DShot ESC setup https://ardupilot.org/copter/docs/common-dshot-escs.html?highlight=blheli

BLHeli 32 passthrough and firmware warning https://en.ardupilot.org/copter/docs/common-blheli32-passthru.html

CRSF and ELRS radio setup https://ardupilot.org/copter/docs/common-rc-systems.html?highlight=sbus

Typical flight controller wiring https://ardupilot.org/copter/docs/common-flight-controller-wiring.html

Wiring and integration notes

Device	Connection concept	Critical check
4S battery	Orqa ESC battery pads through XT60 and low ESR capacitor	Observe polarity, strain relief and insulation. First electronics power up through a smoke stopper.
Orqa ESC	LUX motor outputs M1 to M4 plus ground and any verified telemetry, current and voltage signals	No blind connector use. Build and sign off a pin map first.
ELRS R24 D	One full LUX UART with TX and RX crossed, 5 V and ground	SERIALx_PROTOCOL 23. Never feed VBAT.
M10Q 5883	One UART for GNSS plus I2C SCL and SDA, 5 V and ground	Mount remotely and verify orientation.
Tank Solo	Analog camera video input, LUX SmartAudio UART TX, ground and a verified supply rail	Antenna attached before power. The Pi camera is not connected here.
Analog FPV camera	Tank Solo video input and suitable regulated supply	Use the camera voltage range from its own manual.
Pi Zero 2 W	Dedicated 5 V BEC from 4S; optional MAVLink UART with common ground	Confirm UART voltage level and disable any console on the selected serial port.
Pi Camera 2	CSI ribbon through a Pi Zero 22 pin camera cable	Route away from sharp carbon and propeller arcs.

Mechanical and RF layout

- Place the FC on soft grommets with its orientation recorded. Keep the ESC and high current battery loop compact.
- Mount the M10Q 5883 away from motors, ESC, battery leads, VTX and steel fasteners. A rear mast is preferable if it stays clear of propellers.
- Mount the two ELRS antennas as a true diversity pair with different orientations and clear of carbon. Keep them separated from the 5.8 GHz VTX antenna and Pi Wi Fi antenna area.
- Use two battery straps and a non slip pad. The published frame plus battery plus four motors is already about 1.31 kg before electronics, wiring, antennas and payload.
- Use Kapton or fishpaper at carbon contact points, protect wire exits with braid or spiral wrap, and apply removable blue threadlocker only to clean metal to metal threads.

Safety and first power sequence

1. Props remain off for all firmware, radio, GPS, VTX control, motor order and motor direction work.
2. Continuity check the battery input for a short and verify XT60 polarity before connecting any pack.
3. Use a smoke stopper for the first low load electronics power up. It is not used for motor thrust testing.
4. Attach a rated 5.8 GHz antenna to the VTX before the VTX receives power.
5. Verify every regulated rail with a multimeter before plugging in the FC, receiver, GPS, VTX, analog camera or Pi.
6. Perform failsafe, arming switch, mode switch, GPS, compass and motor order checks before installing props.
7. For the first propeller test, restrain the test article, measure current, monitor motor and ESC temperature, and stop immediately for vibration, desynchronization, excessive current or heat.
8. Charge and store the LiPo with an appropriate balance charger in a fire resistant location. Inspect the pack after every high current test.

Open items from crosscheck

The accompanying workbook is the authoritative item by item crosscheck. It treats every inventory row as available regardless of wishlist or order status, as requested.

Crosscheck state	Item	Handover note
Missing	M5 propeller locknuts	Verify actual motor shaft and thread before purchase.

Crosscheck state	Item	Handover note
Missing	FC to ESC 8 pin harness	The LUX box may include JST SH leads, but an electrically compatible Orqa harness is not established. This remains missing until pin compatibility is verified.
Missing	Kapton tape or fishpaper	Electrical isolation from carbon and sharp edges.
Missing	Foam or 3M VHB tape	Mounting and vibration control.
Missing	Braided sleeve or spiral wrap	Wire abrasion protection.
Missing	Self powered lost model buzzer	Recovery after main battery separation.
Missing	Propeller balancer	Useful for validating large tri blades.
Partial	M3 hardware set	Screws present; standoffs, washers and locking nuts not confirmed.
Partial	Zip tie anchors	Zip ties present; adhesive or screw anchors not confirmed.
Partial	Equipment mounts	ELRS and standard camera mounts present; GNSS, Pi and Pi camera mounts not confirmed.
Verify	Blue removable threadlocker	Generic Loctite appears in inventory; confirm it is removable medium strength, not permanent.
Conditional	FAA approved Remote ID module	Only if required for the planned operation.

Decisions already made

- Use the on hand 4S battery for the current build phase.
- Do not recommend replacements for photographed or inventoried parts merely because another option may be preferable.
- Identify only additional materials, interfaces or safety items needed to make the existing component set work.
- Treat wishlist, ordered and on hand spreadsheet rows as accessible for crosscheck purposes.

Next chat starting point

Upload this DOCX and the XLSX, then paste the following request into the new chat.

> *Continue the 10 inch FPV drone build from the attached handover and component crosscheck. Keep the build on 4S and do not recommend replacements for components already present. First produce a verified pin by pin wiring plan for the Lumenier LUX F765, legacy Orqa 3030 70A BLHeli 32 V2.2 ESC, Matek ELRS R24 D, Matek M10Q 5883, Rush Tank Solo, analog FPV camera and dedicated Pi power regulator. Explicitly resolve the FC to ESC harness pinout before any plug connection. Then create the ArduPilot Mission Planner setup sequence and a restrained propulsion validation plan for the XING2 3110 900KV motors with Gemfan 1050 tri blade props on the Tattu 4S pack. Flag any facts that still require a board photo, continuity test or manual confirmation.*

Information to capture next

- Clear top and bottom photographs of the LUX F765 and Orqa ESC with all pad labels visible.
- The exact 8 pin cables found inside the LUX box and the connector fitted to the Orqa ESC, including wire colors and connector keying.
- Analog FPV camera make, model and voltage range.
- On aircraft 5 V BEC model and rating intended for the Pi.
- Radio transmitter and ELRS module model, firmware major version and regulatory domain.
- Measured motor shaft diameter, prop fit, vehicle all up weight and any bench current data.

Handover limitations

This document consolidates product pages and official project documentation available at preparation time. It does not replace board level pinout confirmation, continuity testing, local flight regulations or controlled propulsion testing. The Rekon10 frame did not yield a standalone manufacturer assembly manual, and an exact standalone manual for the CanaKit adapter was not located; the cited references and photographed labels are used for those items.